

CPG Analytics — Product & UX Discussion

Working document for our product/UX discussion.

We discuss here freely; at each milestone we snapshot with a date/time stamp.

Status

- **Mode:** UX / product discussion (no code yet)
- **Started:** 2026-06-01 16:15 IST
- **Last updated:** 2026-06-08 11:26 IST

Governing Principle (applies to everything below)

Facts are deterministic. Language is intelligent.

Cube.js + SQL + Python compute every number, ranking, threshold, and score.

The LLM only narrates, explains, suggests wording, classifies intent, and converses.

The LLM never invents metrics, never bypasses RBAC, never presents a "possible reason" as proven.

Topic 1 — Knowing the User & Building Memory

The core reframe: two profiles, never confused

There is no single "image of the user." There are two profiles on opposite sides of the governing principle:

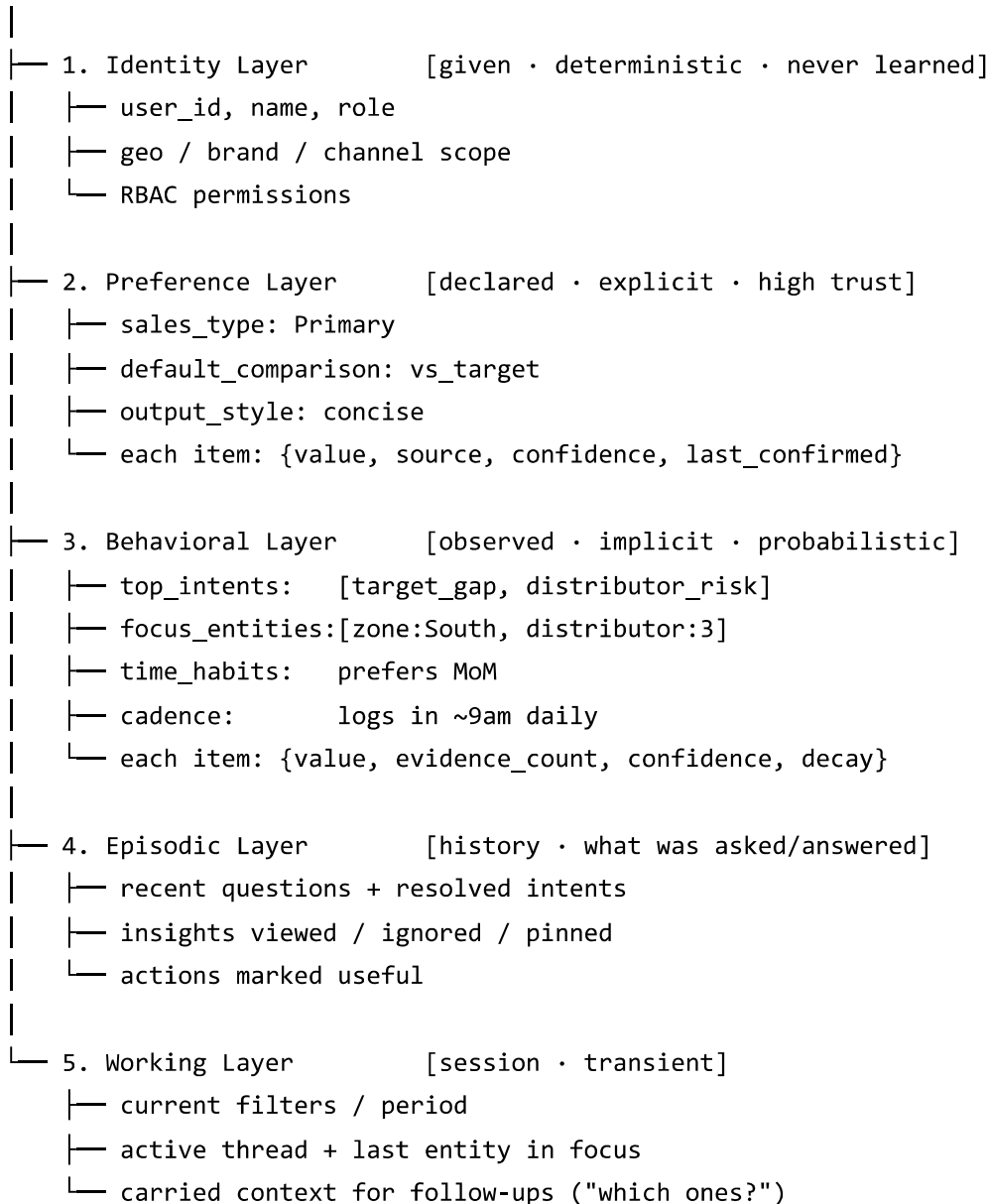
- **Given identity** — role, geography scope, RBAC. Deterministic, from login/HR.
We **never infer or learn** this. (Inferring scope could leak data the user shouldn't see.)
- **Learned model** — preferences, habits, defaults, recurring questions. Probabilistic, from interaction. Always correctable, always reversible.

The seed problem (from the chat screenshot)

User asks "How is my sales doing today" → system asks "Primary or Secondary?".
That's fine the *first* time. By the *tenth* time, if he always picks Primary, we should not be asking anymore. The lifecycle **ask** → **default** → **assume** is the whole game.

The internal "image of the user" (memory model — 5 layers)

User Model (the internal image)



Subsequent interactions — the clarification lifecycle

Clarification lifecycle (per ambiguous slot, e.g. Primary/Secondary)

```

└─ Stage 1  ASK          → n=0, no signal → full question
└─ Stage 2  DEFAULT      → some signal → pre-select last choice,
|                               still show the toggle ("Primary ▼")
└─ Stage 3  ASSUME       → strong, consistent signal → answer silently,
|                               show "(Primary · change)"

```

Correction at any stage → demote confidence, re-learn

Governance guardrails (critical)

- NEVER infer RBAC / scope from behavior.
- Preferences are reversible + visible ("why am I seeing this?").
- Confidence decays if habits change (recency-weighted).
- Learned ≠ locked: always offer the override.

Topic 2 — Make Cold Start Easy: Curated Catalog ("map everything, default most, ask few")

User's idea: don't clarify reactively mid-conversation (repetitive, annoying). Instead pre-curate a **catalog of every disambiguation that could ever come up**, map each to the metrics/values it resolves, then at start ask only the 2-3 highest-value ones. Everything else gets a silent default the catalog already knows.

The Curated Question / Preference Catalog

Preference Catalog (curated once, reused for every user)

- |
- |— Each entry = one "slot" the system might need resolved
 - | |— slot_id e.g. sales_type
 - | |— question_text "Which sales view do you usually work with?"
 - | |— options [Primary, Secondary]
 - | |— maps_to ← THE KEY: what this answer controls
 - | | |— measures primary_net_sales | secondary_net_sales
 - | | |— fact_table sales_fact | secondary_sales_fact
 - | | |— default_unit value | units
 - | |— default_by_role {SO: Secondary, ASM: Secondary, NSM: Primary}
 - | |— ask_priority high | medium | never-ask (just default)
 - | |— resolves_downstream which other slots this answer cascades into
 - | | |— reversible true (always shown as a toggle later)
- |
- |— Slot categories (curate broadly here)
 - | |— Data source Primary vs Secondary · value vs volume
 - | |— Time framing MTD · today · vs last month · vs LY
 - | |— Comparison basis vs target · vs last period · vs LY
 - | |— Granularity focus zone · distributor · SKU · channel · outlet
 - | |— KPI priority net sales · growth% · target ach% · gap value
 - | |— Channel focus General Trade · Modern Trade · E-comm
 - | |— Brand/category focus which brands he tracks
 - | | |— Output style concise · detailed · table · manager-summary
- |
- |— Feeds BOTH
 - | |— Cold-start onboarding (ask the few high-priority ones)
 - | | |— In-chat fallback (if a slot is ever still unknown, ask then)

Cold-start strategy — ask MINIMAL

- Cold Start
- └ Step 1: Resolve everything we can SILENTLY (role defaults + scope)
 - └ Step 2: Score remaining ambiguity per slot
 - | ask_score = ambiguity × impact × frequency
 - └ Step 3: Ask ONLY top 2-3 by ask_score
 - | prefer slots that CASCADE; bundle into one short card (not a quiz)
 - └ Step 4: Everything else = assume + reveal as editable chips
 - | "MTD · South Zone · vs Target ▼"
 - └ Step 5: Write resolved slots to Preference Layer
 - asked → confidence high (declared)
 - defaulted → confidence medium (role prior)

Starter catalog (concrete — "map it with metrics and values")

slot	question (asked only if high- priority)	options → maps to	role default	ask?
sales_type	"Primary or secondary sales?"	Primary→ primary_net_sales / Secondary→ secondary_net_sales	SO/ASM: Secondary · NSM: Primary	ask
measure_unit	"Track by value or volume?"	Value→ net_sales / Units→ net_units	Value	ask
comparison_basis	"Against target or last month?"	→ target_achievement_pct / sales_growth_pct	vs target	maybe
time_framing	—	→ period dimension	MTD	never-ask
granularity_focus	—	→primary drilldown dim	RSM→zone, ASM→distributor	never-ask

slot	question (asked only if high- priority)	options → maps to	role default	ask?
channel_focus	—	→ channel filter	All / by scope	never-ask

Cap cold-start at 2-3 questions, hard limit. Everything else defaults.

Topic 3 — Persona Cards (THE SECRET SAUCE / business model)

We sell the app to CPG companies. The product ships with **vendor-curated persona cards** (SO, ASM, RSM, NSM, Brand Manager, Marketing Manager, Business Analyst). Using deep CPG domain knowledge — the standard **sales hierarchy** and what each role cares about — each card pre-encodes the KPIs, insight types, question bank, and the entire preference catalog with industry-correct defaults.

The buyer does almost no setup: they **assign each employee to a persona** and that person instantly inherits a fully-loaded, correct experience. **Cold start is never cold** — the persona card IS the warm prior. The Preference Catalog from Topic 2 is not built per-buyer; it is pre-loaded inside the persona card.

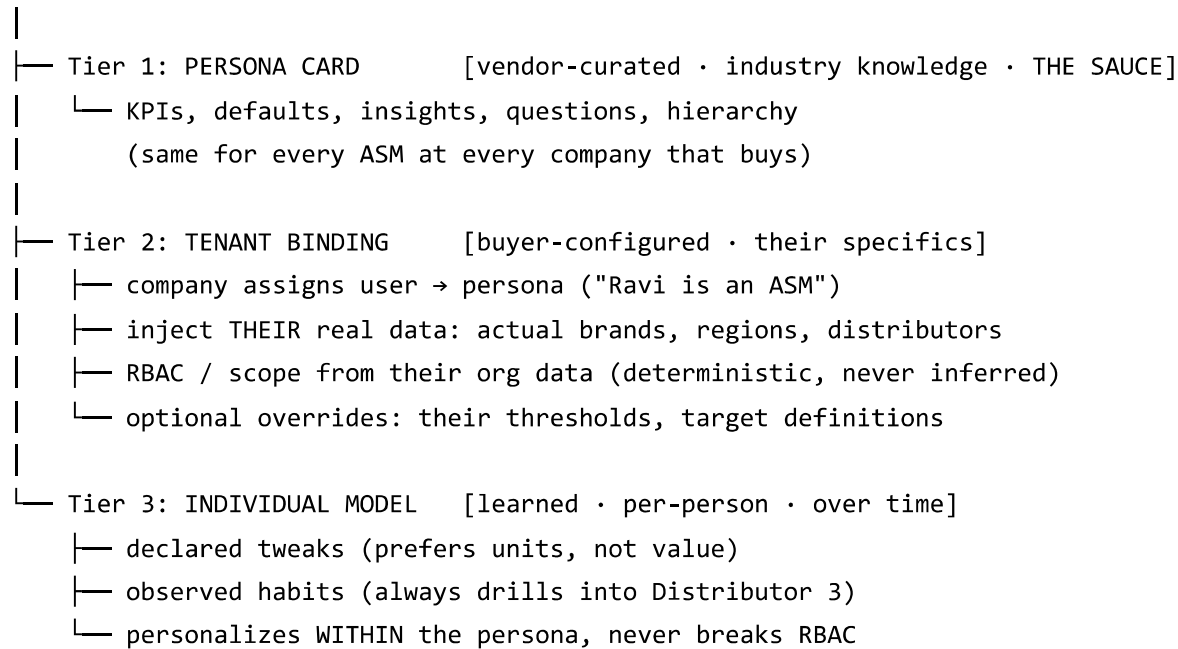
What's inside a Persona Card (vendor-curated)

Persona Card (e.g. "Area Sales Manager") ← shipped with the product

- |
- |— Identity blueprint (what this role IS in CPG)
 - | |— avatar_id: asm
 - | |— hierarchy_position: reports_to → RSM ; manages → SOs
 - | |— typical_scope_shape: a set of distributors in one area
 - | |— responsibility: secondary sales, coverage, distributor health
- |
- |— KPI set (what this role watches)
 - | |— primary: secondary_net_sales, target_achievement_pct, coverage_pct
 - | |— secondary: distributor_stock_days, inactive_outlet_count
- |
- |— Pre-loaded Preference Catalog (defaults, already mapped)
 - | |— sales_type = Secondary · measure_unit = Value
 - | |— granularity = distributor · comparison = vs target
 - | |— every slot pre-filled with the industry-correct value
- |
- |— Insight types that matter to this role
 - | |— distributor_underperformance · outlet_coverage_drop · stockout_risk · target_gap
- |
- |— Curated questions (Ask Analyst + clarification bank)
 - | |— each pre-tagged with intent + entities + cube context
- |
- |— Recommended action style (role-appropriate verbs)
 - | |— "Call distributor" · "Check stock" · "Fix coverage gap"
- |
- |— Minimal ask-set (what to confirm at first login)
 - | |— often ZERO – the card already knows; just reveal + let him edit

Three-tier resolution (how a real user is assembled)

How a real user's experience is assembled



Resolution order: **Persona Card** → **Tenant Binding** → **Individual Model**. Each tier overrides the one above, but only within what's allowed. A user is "an ASM persona, at Company X, who happens to be Ravi."

Consequences

- **Cold start basically disappears** — persona defaults make the 2-3 questions optional.
- **Hierarchy is part of the sauce** — card knows ASM→RSM→NSM, enabling scope roll-up, peer comparison, and escalation ("this is really an RSM-level problem").

Open decisions

1. Persona/tenant boundary — needs a hard line.

- In the card (vendor IP, industry-generic): KPIs, insight types, question bank, default preferences, hierarchy.
- Tenant data (their business): brands, regions, distributors, thresholds, RBAC.
- Getting this boundary right is what lets the same card sell to 50 companies without leaking one company's setup into another.

2. Can a buyer customize a persona card? Three models:

- (a) locked — buyer only assigns users.
- (b) clone-and-tweak — buyer forks the card, adjusts thresholds/questions.
- (c) layered overrides — vendor card stays canonical, buyer adds a thin override layer.

- **Leaning (c):** keeps vendor IP updatable (ship a better ASM card next quarter, everyone benefits) while letting buyers tune.
3. **Cold-start ask aggressiveness** — how boldly to move from "ask" to "assume" (trust vs friction).
 4. **Learned model: per-user only, or also a role-level prior that feeds back?**

Milestones

Milestone 1 — Memory, Curated Catalog & Persona-Card Model — 2026-06-08 11:26 IST

Established the foundational UX/product model for knowing the user:

1. **Two profiles, never confused:** given identity (deterministic, never learned) vs learned model (probabilistic, reversible). 5-layer User Model (Identity / Preference / Behavioral / Episodic / Working).
2. **Curated Catalog approach:** map every disambiguation slot to metrics/values up front; default most from role; ask a hard-capped 2-3 questions at cold start. Clarification lifecycle: ask → default → assume.
3. **Persona Cards = the secret sauce:** vendor pre-curates per-role cards (KPIs, insight types, question bank, pre-loaded preference catalog, hierarchy) from CPG domain knowledge. Buyers just assign users to personas; cold start is never cold.
4. **Three-tier resolution:** Persona Card → Tenant Binding → Individual Model.

Open decisions carried forward: persona/tenant boundary, persona customization model (leaning layered overrides), ask aggressiveness, role-level prior feedback.

Milestone 2 — Moved to Spec Package; Common Persona Card schema drafted — 2026-06-11 13:47 IST

Decided HOW to progress for a coding team: stop producing discussion trees, produce a versioned **spec/ package** of contracts (typed fields + JSON Schema + worked examples + CI validation). `ux.md` = thinking; `spec/` = what the team builds against.

Locked 4 schema-level decisions (v0.1, reversible):

1. One schema per concept, COMMON across all hierarchy levels (differences = field values).
2. Persona Cards are SEMANTIC; layout is a separate common section-grammar.
3. Cards are THIN; they REFERENCE central registries (metrics 12, insight rules 13).

4. Catalog slot definitions are SHARED (11); cards carry only {slot, default, ask_priority}.

Created so far:

- spec/README.md (index, conventions, locked decisions, glossary, build order)
- spec/10-persona-card.schema.md (the anchor — full field table + JSON Schema + validation rules)
- spec/examples/asm.card.yaml + spec/examples/nsm.card.yaml (same shape, different values — proves level-agnosticism)

Still TODO in spec package: 00-architecture.md , 11-catalog.schema.md ,
12-metric-registry.md , 13-insight-rule.schema.md , 14-section-grammar.schema.md ,
20-resolution.md , examples/catalog.yaml .

Milestone 3 — Full spec package complete (DRAFT v0.1) — 2026-06-22 12:06 IST

All spec files are now written and consistent. Ready for team review.

Spec files:

- 00-architecture.md — big-picture on-ramp.
- 10-persona-card.schema.md — common card schema (+ job_description).
- 11-catalog.schema.md — preference/question slots (v0.2, real CPG slots).
- 12-metric-registry.md — metric -> Cube.js measure.
- 13-insight-rule.schema.md — deterministic risk/opportunity/anomaly/driver rules.
- 14-section-grammar.schema.md — home page layout grammar.
- 20-resolution.md — runtime: 3 tiers, RBAC, scope check, geo depth, freshness.

Examples (all cross-referenced and CI-validatable):

- asm.card.yaml , nsm.card.yaml — same shape, different values, with job descriptions.
- catalog.yaml — 9 real slots (sales_type, sales_metric, stock_metric, time_period, history_sales_avg, sales_trend, comparison_period, channel_focus, output_style).
- metric-registry.yaml — 12 metrics.
- insight-rules.yaml — 10 rules across all four types (7 risk, 1 anomaly, 1 opportunity, 1 driver).
- nsm.home.yaml — the NSM home page layout.

Also decided: Q2 (geo depth) — cap by result size, not hierarchy level. See [open-questions.md](#).

Open for the discussion that follows: tenant override model detail, scoring weights, question-bank as a shared library vs on-card, the LLM guardrail/validation design, and which persona to build first.

Milestone 4 — Spec consistency pass — 2026-06-29

Full cross-file audit of the spec package, then fixed every discrepancy found:

1. **Insight-rule count.** Docs claimed 11; the file has 10 (7 risk, 1 anomaly, 1 opportunity, 1 driver). Corrected the count in `ux.md` and `PROJECT_CONTEXT.md`.
2. **ASM operating_grain.** Was `distributor`, which appeared in `no` rule's `applies_to` (so the ASM matched none of its own insight types). Corrected to `area` — matches the role name and makes all four ASM insight types grain-consistent. `rollup_path` trimmed `[area, region] → [region]` (a role no longer rolls up into its own grain).
3. **Actions had no home / two sources of truth.** Rules and cards both encoded insight→action, and several rule-referenced action ids were defined nowhere. Resolved by making the **persona card authoritative**: `card.actions[].from_insight` is the role-appropriate mapping the user sees; `rule.actions` (13) are now **advisory defaults** (field made optional). Added card actions so every authored `insight_type` has a matching role action (ASM: `review_weak_distributors`; NSM: `review_at_risk_distributors`, `review_sku_mix`, `review_stock_cover`). 14 clarified that `action_list` resolves via the card.
4. **Smaller gaps.** Added `chart.trend_grain` to the section grammar (optional; defaults to the `sales_trend` preference); defined the priority buckets `critical|high|medium|low` in 13 (used by 14 `filters.priority`); clarified period precedence in 20 (`scope.default_period` seeds, the `time_period` slot is source of truth); added two card↔rule CI checks to 10 (grain-appropriate insights; every insight actionable).

Known, expected gap (not a defect): only `asm` + `nsm` of the 7 personas are authored, so referential checks that point at unauthored cards (`rsm`, `so`) and their actions stay open until those cards ship.